NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 12: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. In pure inductive inference, given a collection of examples of a target function, the objective is to return a function that approximates the target function. The function is the _____.

A. Utility function
B. Hypothesis
C. Approximation function
D. Classifier
Correct Answer: (B)

Q2. For a Markov Decision Process (MDP), Value Iteration is primarily used to compute the _____ of each state.

A. optimal utility.
B. optimal policy.
C. transition probabilities.
D. shortest path to the goal.
Correct Answer: (A)

Q3. The value of a policy is defined as the expected cumulative sum of _____ obtained while following the policy.

A. Immediate rewards
B. Discounted rewards
C. Maximum rewards
D. Terminal rewards
Correct Answer: (B)

Q4. Which of the following best describes the key advantage of Q-learning over Value Iteration in a reinforcement learning problem?

A. Q-learning directly learns the optimal policy without considering rewards.
B. Q-learning learns state–action values without estimating the transition probabilities.
C. Q-learning requires the complete transition model of the environment a priori.
D. Q-learning learns only the immediate reward associated with each action.
Correct Answer: (B)

Q5. Which statement best distinguishes feature learning in Deep Learning (DL) from traditional Machine Learning (ML)?
A. ML automatically learns hierarchical features, whereas DL uses handcrafted features.

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B. Both traditional ML and DL always require handcrafted features.
C. DL learns features automatically and represents them hierarchically at multiple levels.
D. DL eliminates the need for a learning algorithm.
Correct Answer: (C)

For Q6 – Q10

Reinforcement Learning: Consider a reinforcement learning (RL) agent interacting with its environment through the standard state–action–reward loop. The RL agent may be built using one or more of three core components: policy, value function, and model.

Q6. The "reward hypothesis" that underlies reinforcement learning states that all goals can be described by:

A. minimization of immediate reward.
B. maximization of expected cumulative reward.
C. maximization of cumulative reward.
D. maximization of the discount factor.

Correct Answer: (B)

Q7. Which of the following best defines a "policy" in reinforcement learning?

A. A prediction of future reward for a given state.
B. A map from states to actions that defines the agent's behaviour.
C. A model of how the environment transitions between states.
D. The scalar signal returned by the environment after an action.

Correct Answer: (B)

Q8. An RL agent is in state S and can choose either a_1 or a_2. Suppose that, whenever the agent is in S and chooses a_1, the distribution of the next state and reward is the same regardless of the sequence of states and actions that led to S. Which conclusion is most appropriate?

A. The environment satisfies the Markov property with respect to state S.
B. The environment must be completely deterministic.
C. The agent must use a model-based learning algorithm.
D. The reward must be independent of the action selected.

Correct Answer: (A)

Q9. An agent has estimated

$$Q(S, a_1) = 8,\ Q(S, a_2) = 5.$$

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It chooses a_2 despite $Q(S, a_1) > Q(S, a_2)$. Which of the following best explains this behaviour?

A. Q-learning always chooses the action with the lowest Q-value.
B. The agent may be using an exploration strategy such as ϵ-greedy.
C. Q-values represent immediate rewards only.
D. The environment must be deterministic.

Correct Answer: (B)

Q10. Which of the following best describes the reward function in an RL problem?

A. A mapping of every single state visited by the agent.
B. A function specifying the reward received from states or state-action transitions.
C. The probability distribution governing state transitions and rewards.
D. A fixed schedule of actions the agent follows.

Correct Answer: (B)

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